Hansen and Singleton (1982, 1984) Table 1 Replicate

1. Sample Period: 1959. 02 – 1978. 12

2. Data sources

1) For ND,

- Population (POPTHM), Real personal consumption expenditures: Nondurable goods (chain-type quantity index, DNDGR**A**3M086SBEA), Personal consumption expenditures: Nondurable goods (chain-type price index, DNDGR**G**3M086SBEA) From FRED, Equally & value-weighted return of all NYSE from CRSP

2) For NDS,

Personal Consumption Expenditures: Chain-type Price Index (PCEPI), Personal Consumption Expenditures: Nondurable Goods (PCEND), Personal Consumption Expenditures: Services (PCES), Population (POPTHM), Equally & value-weighted return of all NYSE from CRSP

3. Variables

1) For ND,
Real per-capita nondurable consumption is constructed as

$$C_t^{ND} = \frac{Real\ nondurable\ consumption\ Quantitiy\ index_t}{POPTHM_t}$$

using DNDGRA3M086SBEA and POPTHM.
Gross consumption growth is

$$g_t^{ND} = \frac{\mathrm{c}_{\mathrm{t}}^{ND}}{C_{t-1}^{ND}}$$

Inflation for nondurable consumption is

$$\pi_t^{ND} = \frac{DNDGRG3M086SBEA_t}{DNDGRG3M086SBEA_{t-1}}$$

Real gross stock returns are constructed as

$$R_t = \frac{1 + r_t}{\pi_t}$$

2) For NDS,
Real per-capita NDS consumption is approximated by deflating nominal NDS consumption with the PCE price index and dividing by population:

$$C_t^{ND} = \frac{(PCEND_t + PCES_t)/PCEPI_t}{POPTHM_t}$$

Gross consumption growth is

$$g_t^{NDS} = \frac{\mathrm{c}_{\mathrm{t}}^{NDS}}{C_{t-1}^{NDS}}$$

PCE inflation is

$$\pi_t^{ND} = \frac{PCEPI_t}{PCEPI_{t-1}}$$

Real gross stock returns are

$$R_t = \frac{1 + r_t}{\pi_t}$$

5. Why NDS results is different from original results?
- Data construction must be different but we cannot exactly replicate data construction because real NDS per capita does not be constructed by method of Hansen and Singleton (1982)

6. Results (Using R packages)

INSTRUMENTAL VARIABLE ESTIMATES FOR THE PERIOD 1959:2–1978:12

Cons	Return	NLAG	$\hat{\alpha}$	$\widehat{SE}(\hat{\alpha})$	$\hat{\beta}$	$\widehat{SE}(\hat{\beta})$	χ^2	DF	Prob
NDS	EWR	1	− 0.9360	2.5550	.9930	.0060	5.226	1	.9774
NDS	EWR	2	0.1529	2.3468	.9906	.0056	7.378	3	.9392
NDS	EWR	4	1.2605	2.2669	.9891	.0059	9.146	7	.7577
NDS	EWR	6	0.1209	2.0455	.9928	.0054	14.556	11	.7963
NDS	VWR	1	− 1.0350	1.8765	.9982	.0045	1.071	1	.6993
NDS	VWR	2	0.1426	1.7002	.9965	.0044	3.467	3	.6749
NDS	VWR	4	− 0.0210	1.6525	.9969	.0043	5.718	7	.4270
NDS	VWR	6	− 1.1643	1.5104	.9997	.0041	11.040	11	.5601
ND	EWR	1	− 1.5906	1.0941	.9930	.0034	7.186	1	.9926
ND	EWR	2	− 0.7127	0.9916	.9918	.0034	12.040	3	.9928
ND	EWR	4	− 0.1261	0.8917	.9921	.0035	14.638	7	.9591
ND	EWR	6	− 0.4193	0.8256	.9936	.0033	18.016	11	.9188
ND	VWR	1	− 1.2028	0.7789	.9976	.0027	1.457	1	.7726
ND	VWR	2	− 0.5761	0.7067	.9975	.0027	5.819	3	.8792
ND	VWR	4	− 0.6565	0.6896	.9978	.0027	7.923	7	.6606
ND	VWR	6	− 0.9638	0.6425	.9985	.0027	10.522	11	.5159

Cons	**Return**	**NLAG**	**alpha**	**alpha_se**	**beta**	**beta_se**	**DF**	**chi_sq**	**prob**
NDS	EWR	1	0.789	3.049	0.992	0.008	1	0.744	0.612
NDS	EWR	2	1.583	2.605	0.988	0.007	3	3.315	0.654
NDS	EWR	4	1.851	2.147	0.988	0.006	7	5.248	0.37
NDS	EWR	6	1.683	2.015	0.989	0.006	11	9.997	0.469
NDS	VWR	1	-1.37	2.13	0.999	0.005	1	0.939	0.668
NDS	VWR	2	0.495	1.821	0.996	0.005	3	1.92	0.411
NDS	VWR	4	0.25	1.608	0.996	0.005	7	5.757	0.432
NDS	VWR	6	-0.618	1.522	0.998	0.005	11	10.165	0.484
ND	EWR	1	-1.394	1.11	0.995	0.003	1	2.257	0.867
ND	EWR	2	-0.769	1.009	0.992	0.003	3	7.033	0.929
ND	EWR	4	-0.113	0.871	0.992	0.003	7	12.111	0.903
ND	EWR	6	-0.337	0.825	0.993	0.003	11	16.746	0.884
ND	VWR	1	-1.184	0.747	0.997	0.003	1	1.198	0.726
ND	VWR	2	-0.562	0.687	0.997	0.003	3	4.733	0.808
ND	VWR	4	-0.652	0.657	0.998	0.003	7	9.01	0.748
ND	VWR	6	-1.187	0.6	0.999	0.003	11	10.567	0.52

Note: initial Values -1 and 0.995 for alpha and beta each, Twostep GMM and first W matrix is identical

7. Result (Using Matlab)